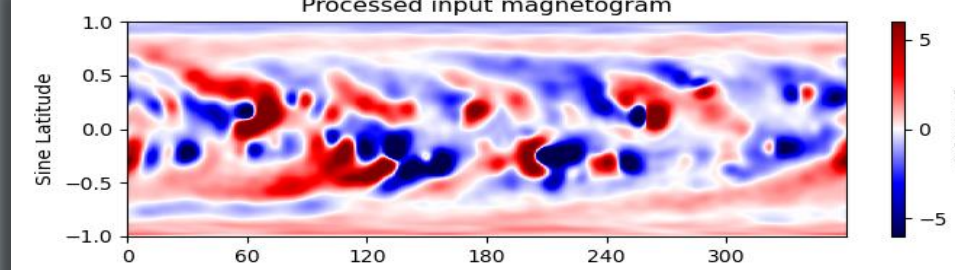
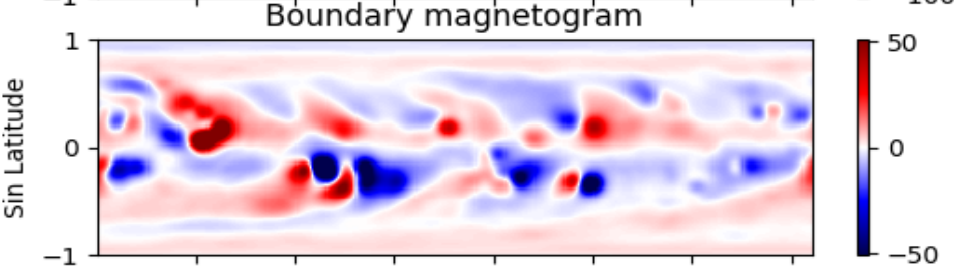
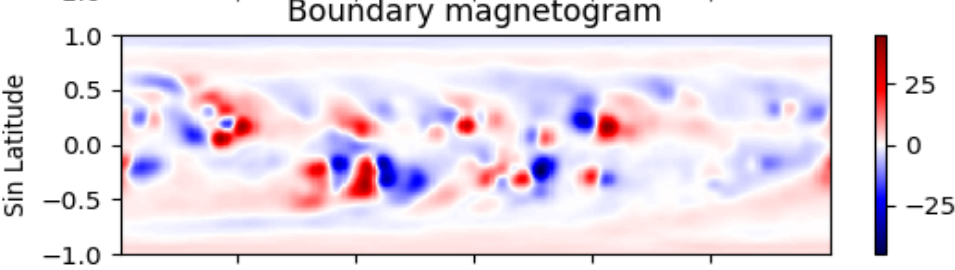
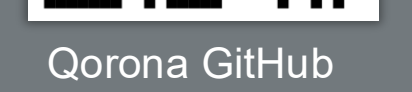
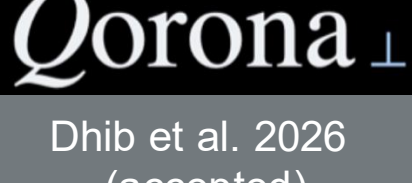


Preparation Phase

Investigated different magnetogram products and processing techniques to identify the best input boundary driving product.



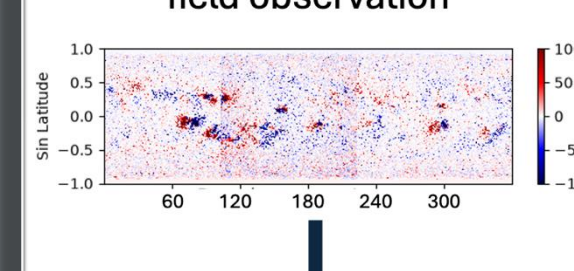
3 type x 2 processing type x 4 $I_{\max}$ filtering $\rightarrow$ 24 types of input maps were investigated in total.



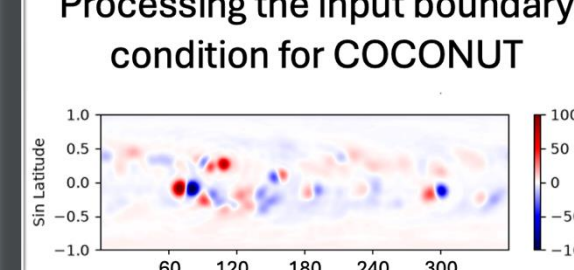
Prediction Phase

Observations

Collecting the latest photospheric magnetic field observation



Processing the input boundary condition for COCONUT



Simulation

Adjust the prediction pipeline to the given day.

Update local and remote storage directories.

Perform COCONUT simulation on the Mindwell cluster of the VSC centrum Flanders.

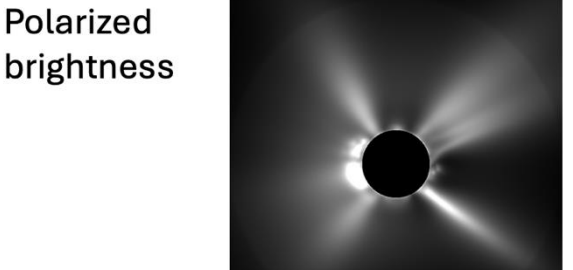
Visualisation

We process the output of COCONUT with Qorona₁ tool.

Squashing factor



Polarized brightness



Publishing

The predictions are automatically

1. Archived in our eclipse prediction database
2. Shared to the live prediction website hosted at ESA
3. Published on the live prediction scientific website hosted at KU Leuven

Steady predictions

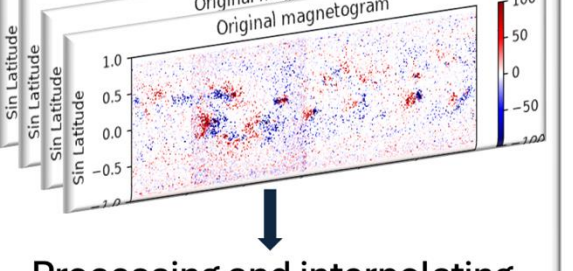
Fully Automated Pipeline

Time - Dependent

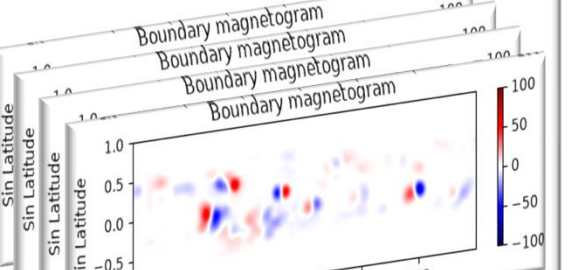
Fully Automated Pipeline

Observations

Collecting all the observations since the last prediction



Processing and interpolating the input boundary conditions



Simulation

Adjust the prediction pipeline to the given day.

Update local and remote storage directories.

Restart COCONUT simulation from the previous solution on the Mindwell cluster of the VSC centrum Flanders.

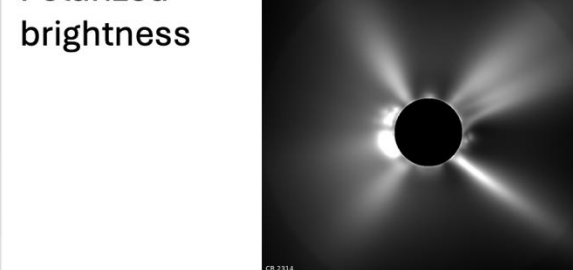
Visualisation

We process the output of COCONUT with Qorona₁ tool.

Squashing factor



Polarized brightness



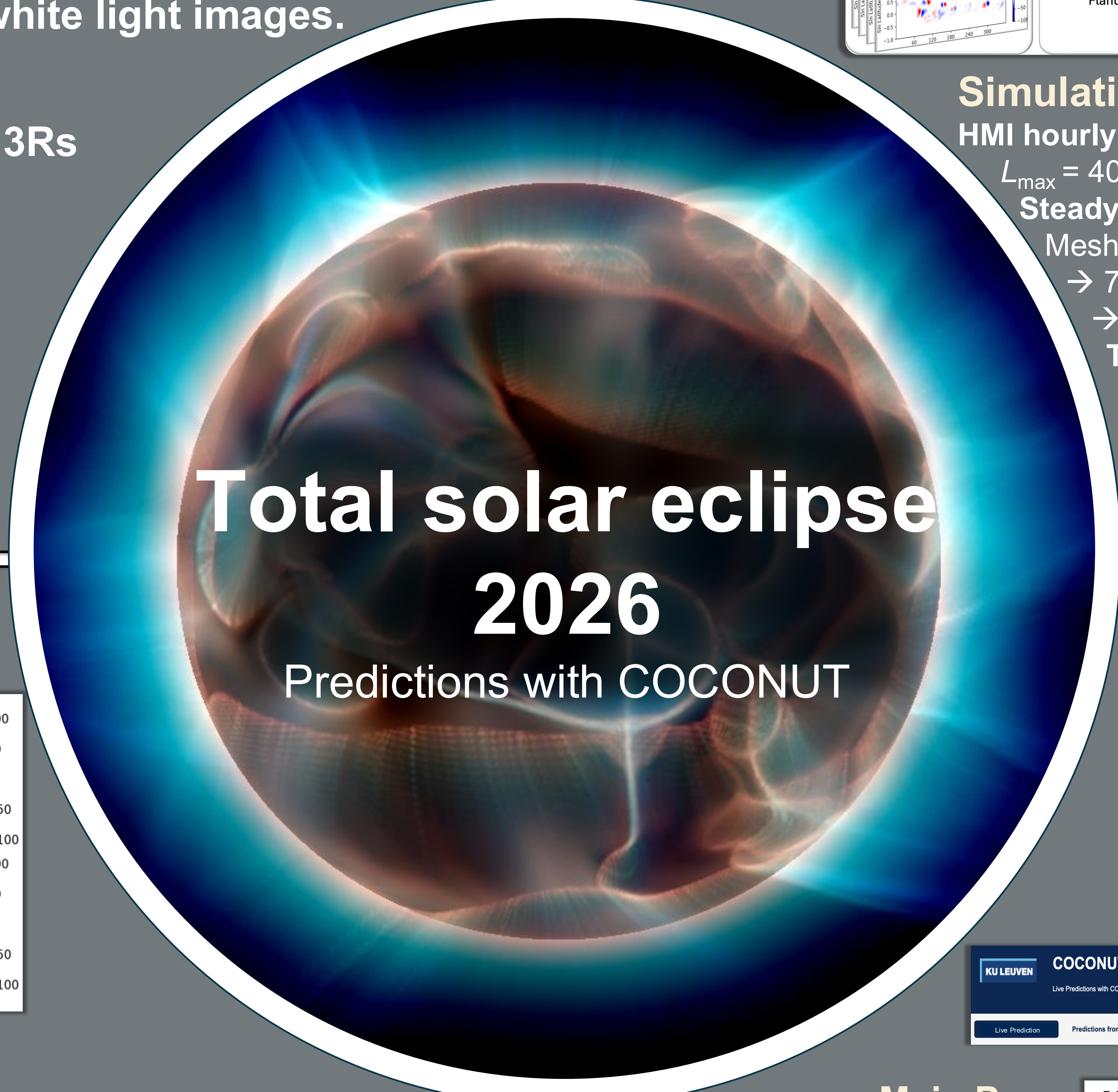
Publishing

The predictions are automatically

1. Archived in our eclipse prediction database
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Reconstructed grid $\rightarrow$ High resolution below 3Rs

Mesh	Surface Triangles	Radial Layers	Total Cells
Level 6 - Default	20,480	73	1.5 M
Level 6 - Refined	20,480	97	1.99 M
Level 7 - Default	81,920	74	6.06 M
Level 7 - Refined	81,920	97	7.95 M



Simulation Setup

HMI hourly input map – high resolution

$L_{\max} = 40 \rightarrow \sim [-60,60]$ G

Steady predictions:

Mesh – Level 7 – enhanced $\rightarrow$ 7.95 M cells

$\rightarrow$ 75% cells inside $< 4R_s$

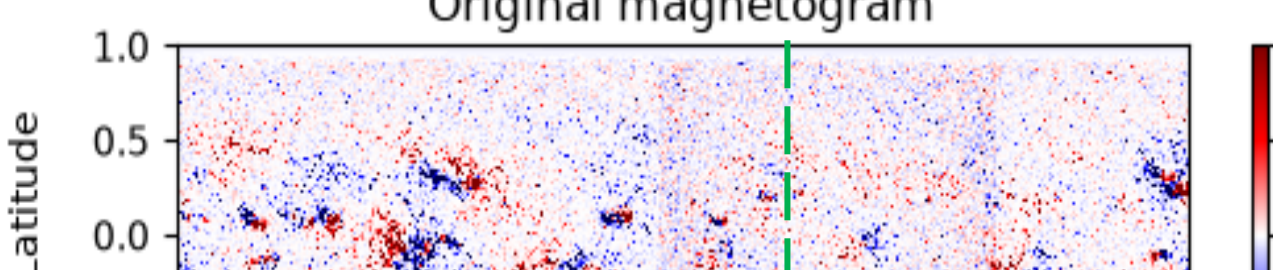
Time dependent predictions

$\rightarrow$ Mesh – Level 6 – enhanced $\rightarrow$ 1.99 M cells

$\rightarrow$ 75% cells inside $< 4R_s$

Results

Input map August 12



Sun-Earth line

Publishing



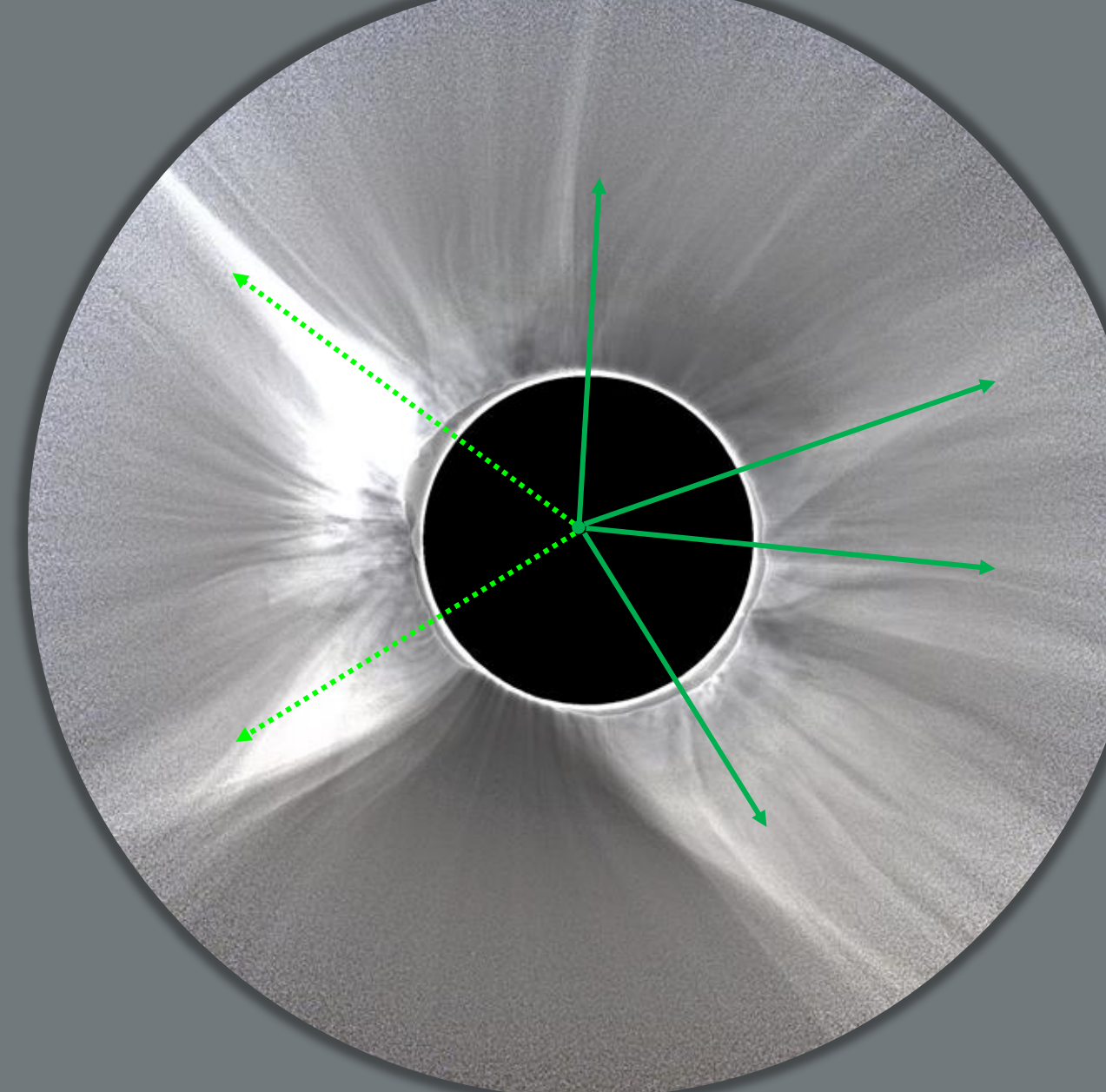
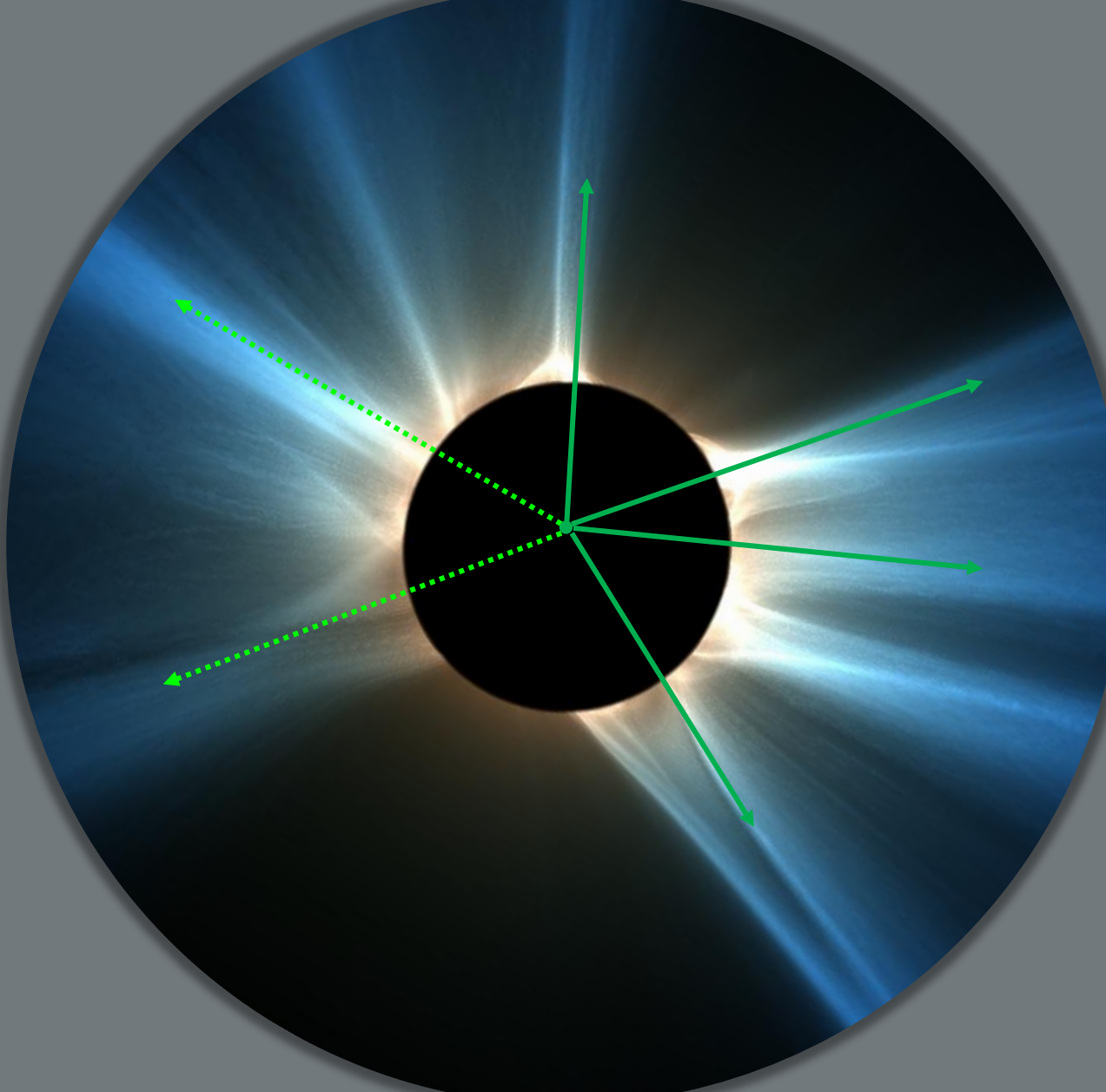
Scientific website



ESA website

Preliminary comparison

There is a global agreement of the coronal structure in the prediction and observations. **Dashed arrows** indicate the slightly misaligned streamers, while **solid arrows** indicate a good agreement.

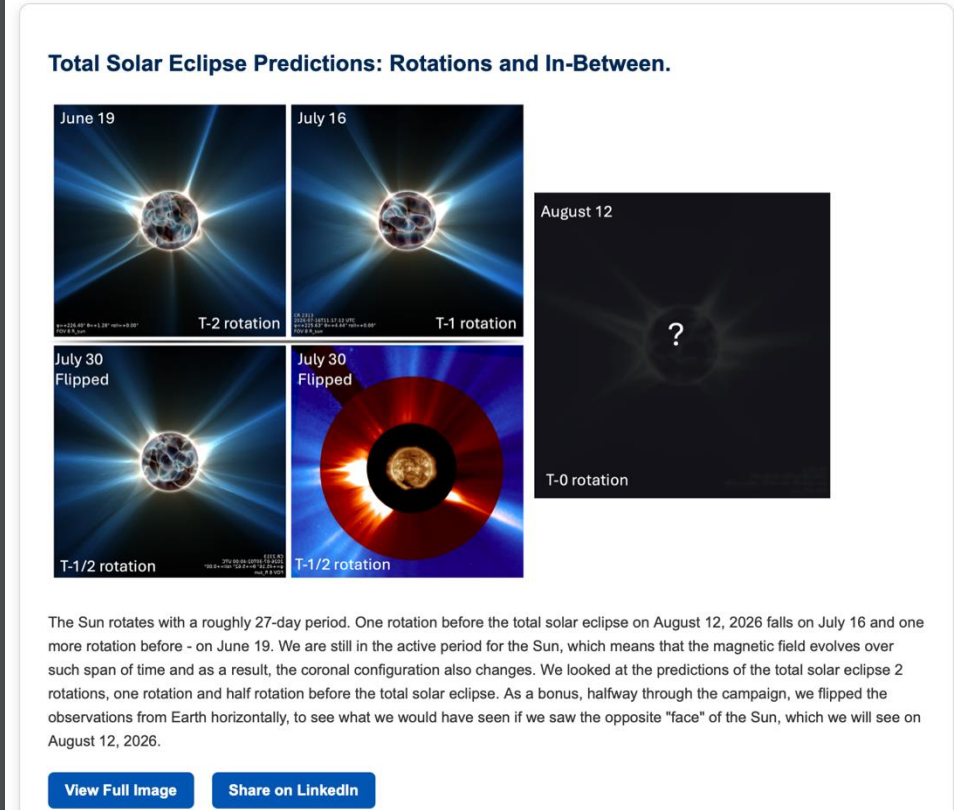
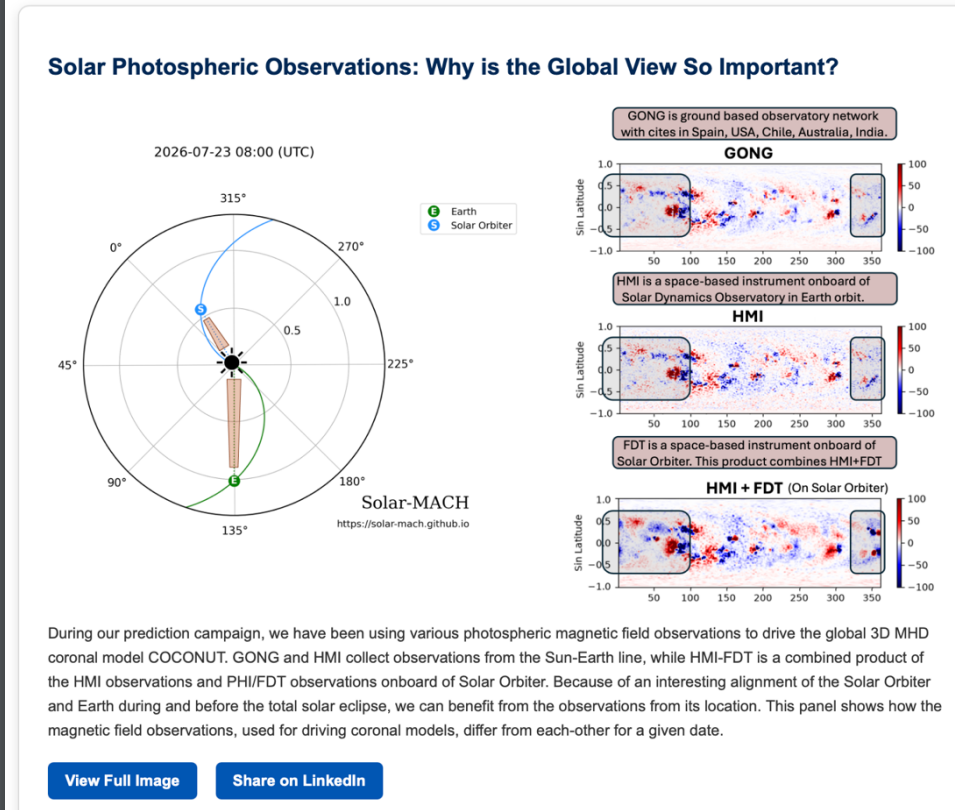


Squashing factor - COCONUT

Image credit: Haopeng Wang, Junyan Liu

Main Page

Live predictions were daily updated on the main page of the website whilst keeping the overall history of daily predictions available on the second page.



Scientific Insights

A dedicated page offers scientific insights within the prediction window.